

RHISHIKESH CHIKHALKAR

SOFTWARE ENGINEER

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SUMMARY

Backend Engineer with 4.5+ years of experience designing, developing, and optimizing scalable backend systems using Python, FastAPI, Django, and PostgreSQL. Proven expertise in microservices, API development, cloud-native applications on AWS, CI/CD automation, and production support. Strong track record of improving application performance, automating data workflows, and delivering secure, reliable solutions in enterprise environments.

PROFESSIONAL EXPERIENCE

Infosys Limited | Dec 2021 - Present | Pune, India

Digital Specialist Engineer

- Designed, developed, and maintained **FastAPI-based microservices** for financial portfolio analytics, serving market and benchmark performance insights.
- Architected and integrated **RESTful APIs** with third-party financial data providers, enabling automated portfolio metric calculations and reporting.
- Optimized **PostgreSQL queries**, indexing strategies, and database schemas, improving API response times by 40% for large-scale financial datasets.
- Implemented **OAuth2-based authentication and authorization** mechanisms, ensuring secure API access and standardized integration across multiple teams.
- Designed and maintained **ETL pipelines** using Python, Pandas, and SQLAlchemy to automate data ingestion, transformation, and reporting workflows.
- Migrated legacy **MFT-based data processing solutions** to **AWS S3**, improving operational reliability, automation, and maintainability.
- Containerized backend services using **Docker** and deployed applications on **AWS EKS Kubernetes clusters**, enabling scalable and resilient cloud-native deployments.
- Built and maintained **CI/CD pipelines** using AWS CodePipeline and GitHub Actions, automating build, testing, and deployment processes.
- Utilized **CloudWatch**, Lens, and Aptakube for production monitoring, log analysis, debugging, and incident investigation, reducing **issue resolution time by 30%**.
- Developed comprehensive pytest-based unit and integration test suites covering APIs, business logic, and PostgreSQL data layers, improving release confidence.
- Enforced automated quality gates within CI/CD pipelines, preventing deployment of untested or unstable code into production environments.
- Conducted **code reviews** and established FastAPI and PostgreSQL development best practices, improving maintainability and reducing review turnaround time.
- Collaborated closely with Product, QA, DevOps, and business stakeholders to deliver fault-tolerant, highly available, and secure backend systems.
- Mentored junior engineers on **API development**, database optimization, debugging techniques, and backend engineering best practices.
- Contributed to **technical documentation**, architecture discussions, and knowledge-sharing initiatives across engineering teams.

TECHNICAL SKILLS

Programming Languages

- Python, SQL

Frameworks & Libraries

- FastAPI, Django, Flask
- SQLAlchemy, Pandas, NumPy

Databases & Caching

- PostgreSQL, MySQL, Redis

Cloud & DevOps

- AWS (S3, Athena, EKS, ECR, CodePipeline)
- Docker, Kubernetes
- GitHub Actions, CI/CD

Testing

- Pytest, unittest, Integration Testing

Tools & Platforms

- Databricks, Linux/Unix
- Lens, Aptakube, Postman

CORE COMPETENCIES

- RESTful API Development, Microservices Architecture, Backend System Design
- Database Design, Query Optimization, Data Integration
- Cloud-Native Applications, Containerization, CI/CD Automation
- Performance Engineering, Scalability Optimization, Production Support
- Secure API Design, OAuth2 Authentication, System Monitoring
- Unit & Integration Testing, Code Review, Technical Mentorship

EDUCATION

Bachelor of Engineering in Computer Science

Shivaji University | June 2016 - June 2020

CERTIFICATIONS / TRAINING

- Google Certified Engineer – Associate
- Infosys Certified Applied Generative AI Professional
- Python for Data Engineering & API Development (Internal Training)

AWARDS / RECOGNITION

- **Platinum Club Member – Infosys Ltd (FY 2025–26)**

Recognized among the top 3% of employees across Infosys for consistent high performance, leadership, and technical excellence. Selected for a year-long professional development and networking program.